

Yuanfan Chen

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EDUCATION

Cornell Tech, Cornell University

Master of Engineering in Computer Science

New York, NY

Aug. 2025 – May. 2026

University of Toronto

Honours Bachelor of Science, Computer Science Specialist (GPA: 3.87/4.0)

Toronto, Canada

Sep. 2020 – May. 2024

EXPERIENCE

Systems Researcher

Far Data Lab, University of Toronto

Jan 2024 – Feb 2025

Toronto, CA

- Co-designed **dpBento**, a modular benchmark framework that standardizes performance evaluation across heterogeneous DPUs (**BlueField-2/3**, **OCTEON TX2**), from micro-level primitives to end-to-end cloud database offloading.
- Profiled **RocksDB** compaction and decompression paths end-to-end; offloaded compute-heavy stages to DPU, reducing **host CPU utilization** by $\sim 40\%$ and achieving $\sim 3\times$ **decompression throughput** via hardware acceleration.

Infrastructure Engineer

Tencent

May 2025 – Sep 2025

Beijing, CN

- Designed **admission control** with **batch-size stepwise profiling** for **TCHouse-X** (ClickHouse-based OLAP) and **SLA-aware priority scheduling** for the LLM serving platform; maintained **100% goodput** under **2–10 $\times$ QPS bursts**.
- Optimized node recovery via fault-tolerant **KV delta-apply**, reducing recovery latency from **10s to 3s** and ensuring data consistency in high-churn distributed clusters.
- Implemented **Continuous Batching** with **Chunk Prefill**; optimized **KV-Cache management** with **PagedAttention** and eviction policies to maximize cache reuse and prevent OOM under high concurrency.
- Optimized CUDA attention kernels for **GQA** (A100) and **DeepSeek FlashMLA** (H200); profiled with **Nsight Compute**, redesigned tiling and warp scheduling; achieved $\sim 20\%$ and $\sim 15\%$ throughput gains respectively.

ML Systems Researcher

Cornell University

Sep 2025 – Present

New York, NY

- Developed **PTA** (Prefill $\rightarrow$ Think $\rightarrow$ Answer) three-stage disaggregated serving with **differential SLO constraints**; configured **DVFS frequency**, **KV-Cache allocation**, and **parallelism** per stage — preserving **TTFAT** via Answer-phase frequency protection while improving throughput.
- Built a **GPU frequency–performance model** via **DVFS** profiling on A100/H100; designed an **online optimization framework** jointly scheduling **LC/Flex/BE** classes under **power cap** and **energy budget** constraints.
- Applied γ -**Boost Queuing Theory** to design a prediction-free, tail-aware scheduler with **KV-cache-aware preemption**; reduced **tail TTFT/TTLT** by **37–60%** while improving throughput by **1.01–1.12 $\times$** on Azure production traces.

TECHNICAL SKILLS

Programming: Python, C++, CUDA, Java **Databases:** RocksDB, ClickHouse, MySQL, Redis

Distributed: Raft, HDFS, fault tolerance, consensus **Networking:** epoll, multiplexing, async I/O, KCP

Systems: Linux internals, CPU/GPU arch, Nsys/Nsight **AI Infra:** vLLM, SGLang, KV-Cache

PUBLICATIONS

- Making Sense of DPU Performance for Cloud Data Processing: [Experiment, Analysis & Benchmark]**. Jiasheng Hu, Chihan Cui, **Yuanfan Chen**, Philip A. Bernstein, Jialin Li, Qizhen Zhang. arXiv:2504.05536 (2025).